

Aliah University

Odd (Autumn)-Semester Examination - 2024

(For 7th Semester BTech(CSE) Programme)

Paper Name: Internet of Things

Paper Code: CSEUGPE12

Full Marks: 80

Time: 3 Hrs

Group - A

(5X2=10)

1. Give a one-sentence answer -

Remember, CO1-CO5

- a) What is a communication protocol widely used in IoT devices for constrained networks?
- b) What functions do sensors and actuators perform, respectively?
- c) What type of computing is required in an IoT system for real-time data processing closer to the devices?
- d) Which programming language is primarily used with Arduino?
- e) What is the primary operating system used by Raspberry Pi?

Group - B

(Answer any 6 questions)

(6X5=30)

- 2. Explain the key components of an IoT system with suitable examples. **Understand, CO1**
- 3. Differentiate between Sensing and Actuation. Provide real-life use cases for both. **Understand, CO1**
- 4. What is Machine-to-Machine (M2M) communication? Discuss its role in IoT. 2+3 **Understand, CO1**
- 5. Write a short note on any two communication protocols used in IoT. **Understand, CO1**
- 6. Describe the basic steps to integrate a temperature sensor with Arduino. Include the circuit setup and code snippet. **Apply, CO2**
- 7. Explain the concept of Interoperability in IoT with an example. Why is it important? 2+3 **Understand, CO3**
- 8. Discuss the role of Python programming in implementing IoT solutions. **Understand, CO3**

Group - C

(Answer any 4 questions)

(10X4=40)

9. a) Explain the architecture of an IoT-based smart home system. **5+5 Understand, CO4**
b) Discuss how fog computing enhances IoT applications. **5+5 Apply, CO2**
10. Write an Arduino code for a smart home application which will constitute of a Heater, Fan and Air Conditioner. Describe the code in details. **5+5 Apply, CO2**
11. Discuss the applications of IoT in the following fields:
a) Healthcare **5+5 Understand, CO5**
b) Agriculture **5+5 Understand, CO5**
12. Write a detailed note on the role of IoT in Smart Cities. Discuss any two use cases. **5+5 Understand, CO5**
13. a) Compare Cloud Computing and Fog Computing in the context of IoT. **5+5 Understand, CO4**
b) Explain the concept of the smart grid and its implementation in IoT. **5+5 Understand, CO4**
14. Propose an IoT solution for monitoring physical activity in elderly individuals. Include system architecture, components, and potential benefits. **5+5 Apply, CO5**